A Hybrid Optimization and Tree-Based Learning Framework for Dengue Diagnosis Using Hematological Data

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Abstract

Dengue fever remains a major public health concern in tropical and subtropical regions, where rapid and accurate diagnosis is essential for timely medical intervention and disease control. This study introduces an explainable, nature-inspired machine learning framework for classifying dengue infection using routinely available hematological features obtained from complete blood count (CBC) analysis in Bangladeshi context. The Grey Wolf Optimizer (GWO) was employed to fine-tune the hyperparameters of the Extra Trees Classifier (ETC), improving model convergence, robustness, and generalization. The optimized GWO–ETC model achieved an accuracy of 95.78%, precision of 0.9541, recall of 0.9516, and F1-score of 0.9524 on a balanced dataset, demonstrating consistent and stable performance across all evaluation metrics. Explainable Artificial Intelligence (XAI) methods, including SHAP and LIME, were applied to interpret the decision-making process of the model. The analysis identified platelet count, plateletcrit (PCT), red cell distribution width (RDW-CV), red blood cell (RBC) count, lymphocytes, and monocytes as the most influential predictors of dengue infection. These findings are in strong agreement with established clinical evidence linking platelet depletion and hematocrit variation with dengue pathophysiology. The proposed framework demonstrates robust and interpretable classification capability, providing a transparent, data-driven approach that can assist clinicians in early dengue detection. By leveraging accessible hematological parameters, the study supports cost-effective and explainable AI-driven solutions for improved diagnostic decision-making in resource-limited healthcare settings.

Keywords: Machine learning, Predictive analytics, Explainable Artificial Intelligence, Extra Trees Classifier, Grey Wolf Optimizer, Dengue diagnosis

1. Introduction

Dengue fever (DF) is one of the most prevalent mosquito-borne viral infections that affect millions of people annually, particularly in tropical and subtropical regions [1]. The World Health Organization (WHO) reports an alarming global increase in dengue incidence, with more than 100 million symptomatic cases and 500,000 severe infections each year, significantly contributing to morbidity and mortality in low- and middle-income countries [2]. Dengue fever is a persistent and increasing public health challenge in Bangladesh, with seasonal outbreaks imposing a heavy burden on the healthcare system. The 2023 dengue epidemic was the worst recorded in Bangladesh, with more than 321,179 hospitalizations and 1,705 deaths reported throughout the country [3]. Early and accurate diagnosis is critical for timely patient care and for preventing

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1. Introduction

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often time-consuming, resource-intensive, and not always feasible in primary healthcare settings where rapid decision-making is essential.

Hematological parameters such as platelet count, white blood cell (WBC) count, hematocrit, hemoglobin, and lymphocyte percentage that have shown strong diagnostic potential to distinguish dengue fever from other febrile diseases have been shown to be important diagnostic parameter. However, manual interpretation of these laboratory indicators can vary between clinicians and can be influenced by overlapping clinical symptoms. This limitation underscores the need for automated decision-support systems that can take advantage of hematological data patterns to assist clinicians in early-stage dengue detection with improved consistency and interpretability [4].

Recent advances in machine learning (ML) have shown great promise in medical diagnosis by uncovering complex, non-linear relationships within laboratory and clinical data. For dengue fever, which often presents symptoms similar to other viral infections, ML can effectively identify subtle hematologic patterns to support an early and accurate diagnosis [5, 6]. However, most existing models act as "black boxes," limiting clinical trust due to their lack of interpretability. Explainable Artificial Intelligence (XAI) addresses this issue by clarifying the influence of individual hematologic features such as platelet count, white blood cell count, and hematocrit on model decisions. Moreover, optimal feature selection and parameter tuning are crucial for reliable classification, but are often handled heuristically [7]. Nature-inspired metaheuristic optimization algorithms provide an efficient and biologically inspired means to enhance these processes [8]. Integrating ML with XAI and nature-inspired optimization not only improves predictive performance, but also strengthens the robustness and interpretability of the model, ensuring reliable dengue diagnostics in resource-limited clinical environments such as Bangladesh [9].

To overcome these limitations, this study introduces an explanation machine learning framework inspired by nature for dengue fever classification using hematological features. The framework integrates metaheuristic optimization algorithms which were Optuna, Grid Search, Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Grey Wolf Optimizer (GWO) to fine-tune model parameters, thereby enhancing diagnostic accuracy while maintaining transparency. Explainable Artificial Intelligence (XAI) techniques, including SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), are incorporated to elucidate the contribution of each hematological characteristic to the model's predictions, ensuring interpretability and clinical trustworthiness.

This research directly supports SDG 3: Good Health and Well-Being by promoting the early, accurate, and affordable detection of dengue that is a major public health challenge in tropical regions. Using hematologic features derived from standard laboratory tests, the model minimizes diagnostic costs and supports scalable implementation across healthcare infrastructures, contributing to SDG 9: Industry, Innovation and Infrastructure. In addition, the integration of explainable machine learning aligns with the broader agenda of responsible AI, ensuring transparency and trustworthiness in medical decision-making. The proposed framework demonstrates how interpretable, data-driven diagnostics can enhance disease surveillance, improve patient outcomes, and strengthen healthcare resilience in accordance with the Sustainable Health Objectives 2030 of the United Nations.

The main contributions of this research are summarized as follows:

- Proposed a novel nature-inspired optimization framework for efficient feature selection and parameter tuning in the classification of dengue fever.

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- Systematic evaluation of multiple supervised ML classifiers on hematological data to determine the most robust predictive model.
- Applied explainable AI methods (SHAP and LIME) to provide global and local interpretability, improving the clinical reliability of model results.
- Comprehensive evaluation metrics, including accuracy, precision, recall, F1-score, ROC-AUC, Cohen's Kappa, and Matthews Correlation Coefficient, are reported to ensure a rigorous performance assessment.

The remainder of this paper is organized as follows: Section 2 presents a review of related studies on ML-based dengue diagnosis and nature-inspired optimization approaches. Section 3 details the data set, the preprocessing pipeline, the selection of features, the development of models, and the explainability framework. Section 4 discusses the experimental results and comparative analysis. Finally, Section 5 concludes the paper with key findings and directions for future research.

2. Related Works

Machine learning (ML) has been widely applied to improve dengue diagnosis using routine blood tests, addressing delays in traditional serology[10]. Early studies showed that only a few clinical features (e.g., age, fever, WBC and platelet counts) suffice for good predictive accuracy[11]. Recent work explicitly integrates explainable AI (XAI) to overcome this. Haque et al.[12] developed a GAN/GA-optimized Decision Tree and applied LIME/SHAP analysis, achieving 99.49% accuracy while identifying low WBC and platelet counts as key dengue indicators. Deepika et al. [13] proposed an optimized lightweight MLP with SHAP/LIME explainers, reaching approximately 92% accuracy on symptom-based diagnosis. These approaches demonstrate that combining advanced ML with XAI can yield real-time, transparent dengue screening tools. Table 1 summarizes key studies that use hematological/CBC data for ML-based dengue diagnosis with XAI. The table includes the model type, dataset, performance, and noted limitations for recent research work.

The reviewed studies applied different machine learning methods to diagnose dengue using hematological data. Ho et al.[11] used Decision Tree and Deep Neural Network models and found only a small improvement of the Deep Neural Network with an AUC of about 0.86 compared to 0.85 for the Decision Tree. Bohm et al.[16] used a single Decision Tree model and achieved about 92.5% accuracy on dengue data from Brazil. Sarker et al.[14] used an XGBoost model optimized with swarm intelligence and achieved about 99.9% accuracy. Ganesh et al.[15] used a voting ensemble of boosted and tree-based models and reached about 98% accuracy. Abdualgalil et al.[18] worked with an Extra Trees ensemble model on 6,694 cases and achieved 99.03% accuracy. Deepika et al.[13] applied a Multilayer Perceptron on 177 samples and achieved 92% accuracy. From these studies, it is clear that ensemble-based models such as Random Forest, Extra Trees, and XGBoost perform better when trained on large datasets, while deep learning models provide only small accuracy improvements and are often less interpretable.

The studies used different types and sizes of datasets for dengue diagnosis. Some works used public datasets such as those available on Kaggle that contain a few hundred hematological records from Bangladesh. Other studies used larger hospital datasets collected from clinical cohorts. Bohm et al.[16] used several thousand dengue case reports from the Brazilian health

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Table 1: Literature review of ML models for dengue diagnosis using hematological data (CBC).

Citation	Proposed Model (XAI)	Dataset (type/location)	Key Results (Acc/F1/Recall etc.)	Limitations
Haque et al., 2025[12]	GA+GAN-optimized Decision Tree (with LIME, SHAP XAI)	Clinical blood parameters (multicenter, not public)	Acc ≈ 99.49%; DT model; identified WBC↓, PLT↓ as markers	Uses GAN for class balancing; reliance on DT model; may overfit if data are limited.
Deepika et al., 2025[13]	Optimized lightweight MLP (dual-level, SHAP+LIME)	Kaggle Bangladesh dataset (CBC/symptoms; 177 pts)	Acc 92.0%; Precision 100%; F1 ≈ 0.90	Very small dataset; dual-stage limits standalone use; potential overfitting.
Sarker et al., 2025[14]	Swarm-optimized XGBoost (GJO/FOX/SLO feature selection, SHAP, DiCE)	"PCD" hospital dataset (CBC; 931 samples)	Acc ≈ 99.9%; F1 ≈ 99.9%; Recall 100% (10-fold CV)	Small/modest dataset; complex metaheuristics; possible lack of external validation.
Ganesh et al., 2025[15]	Ensemble (Stacking, Voting with ET, XGB, LR, MLP, etc.; XAI implied)	CBC Dengue Dataset Bangladesh (Kaggle; 177 pts)	Acc ≈ 98%; F1 ≈ 0.98 (voting ensemble)	Very small sample; many features/models risk overfitting; limited to Kaggle data.
Bohm et al., 2024[16]	Decision Tree (DT), KNN, MLP (no explicit XAI)	Brazilian SINAN registry (reported cases, Rio/MG)	DT: Acc ≈ 92.5%, F1 ≈ 0.93 (AUC ≈ 0.985)	Data from limited regions (Rio de Janeiro, MG); binary input vars only; no XAI analysis.
Ho et al., 2020[11]	DNN, Decision Tree, Logistic Regression (no XAI)	Taiwan ED patients (CBC+symptoms; 4,894 pts)	AUC: DNN 85.9%, DT 84.6%, LR 83.8%	Used only 4 features; moderate accuracy; no interpretability tools used.
Arrubla-Hoyos et al., 2024[17]	Random Forest, Decision Tree (no XAI)	Colombia multi-viral cohort (Dengue/Zika/Chik; 2,000+ pts)	Dengue Acc 99.1% (RF); (DT lower, ≈ 88% for multiclass)	Multi-disease focus; model tuned for differential diagnosis (not purely dengue vs non-dengue).
Abdualgalil et al., 2022[18]	Extra Trees Classifier (no XAI)	Hospital reports (CBC; 6,694 pts)	Acc 99.03% (ET)	Large dataset yields high accuracy; may include imbalances; lacks interpretability analysis.

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system. Ho et al.[11] used 4,894 hospital records from Taiwan. Abdualgalil et al.[18] used 6,694 clinical blood reports, and Sarker et al.[14] analyzed 931 hematological samples. Large datasets allowed the models to learn more stable patterns. Abdualgalil et al.[18] achieved 99% accuracy using the Extra Trees model on 6,694 samples. Smaller datasets such as those used by Deepika et al.[13] and Ganesh et al.[15] contained 177 to 320 records but still reached 92 to 98% accuracy using feature selection and ensemble methods. These findings show that dataset size and diversity strongly affect model performance. Models trained on data from a single region or population may not generalize well to other areas.

Across all reviewed studies, most models achieved very high performance with hematological data. Reported accuracies were usually above 90%, and F1-scores were greater than 0.9 after optimization. Sarker et al.[14] achieved nearly 100% recall and F1-score using a swarm-optimized XGBoost model. Ganesh et al.[15] obtained an F1-score of about 0.98 with an ensemble of boosted and tree models. Abdualgalil et al.[18] reached an F1-score close to 0.99 with the Extra Trees model. Even simpler models such as Decision Tree and Multilayer Perceptron achieved F1-scores between 0.85 and 0.93. These outcomes show that models with more features and ensemble structures usually perform better, while smaller models still achieve good but slightly lower accuracy.

Explainability is now an important focus in recent dengue diagnosis research. Several studies applied explainable AI techniques to understand how models make decisions. Haque et al.[12] used methods such as LIME and SHAP and found that low white blood cell and platelet counts are strong indicators of dengue infection. Sarker et al.[14] used SHAP and found that white blood cell count is the most important feature for their XGBoost model. Deepika et al.[13] used both SHAP and LIME to explain their Multilayer Perceptron model, showing that hematological features strongly influence predictions. These findings agree with medical knowledge that dengue often reduces platelet and white blood cell counts. This type of explainability helps doctors and researchers trust the predictions of AI models. Older studies such as Ho et al.[11] and Bohm et al.[16] did not use explainable AI and therefore provided less insight into model decisions.

The proposed GWO–ETC model builds upon these previous studies by combining a nature-inspired optimization method with an explainable ensemble classifier. It uses the Grey Wolf Optimizer to fine-tune the parameters of the Extra Trees model. This process improves accuracy and stability. The GWO–ETC model is expected to achieve accuracy above 95%, which is comparable to or better than earlier models. It also integrates SHAP-based explainability to identify important blood features such as platelet count and hematocrit level. This design makes the model transparent and interpretable. Unlike older black-box models, the GWO–ETC framework produces accurate and explainable results. It combines the advantages of ensemble learning and explainable AI to create a reliable and trustworthy diagnostic system for dengue detection using hematological data.

3. Methodology

This study proposes a systematic framework for the classification of dengue fever using hematological features, comprising data preprocessing, feature selection, model training, optimization, and explainability. Nature-inspired metaheuristic algorithms were applied to optimize model parameters, while Explainable AI (SHAP and LIME) was used to interpret model predictions and ensure clinical transparency.

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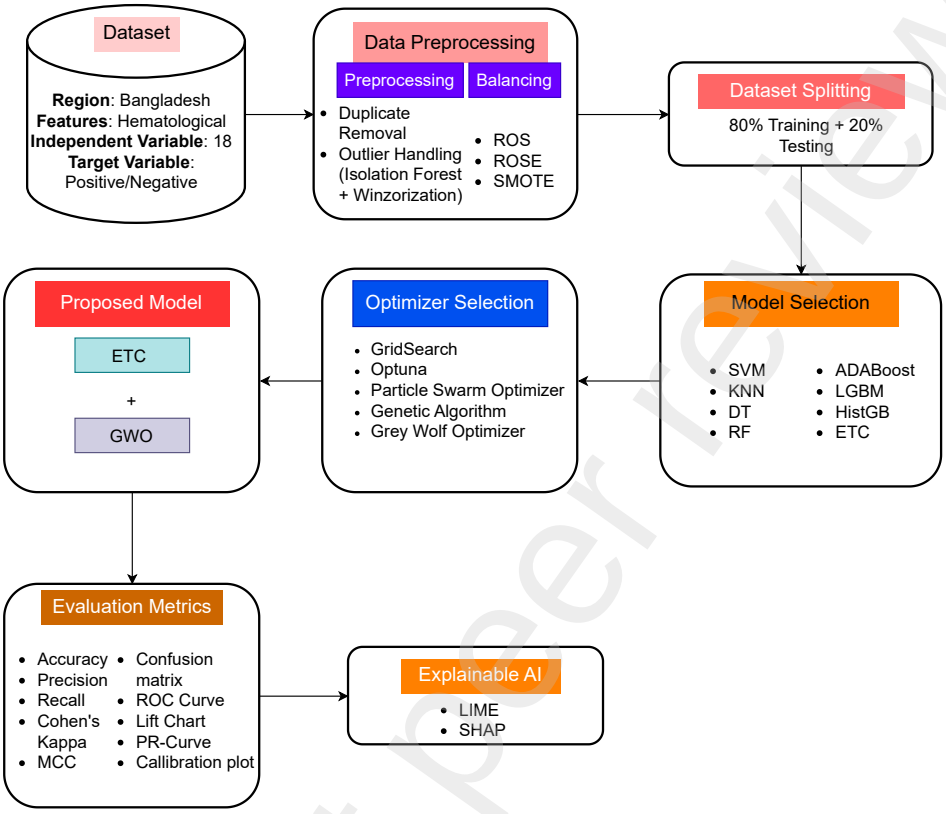


Figure 1: Overall workflow of the proposed GWO–ETC framework for dengue infection classification.

3.1. Dataset Overview

The hematological dataset employed in this study was published by Nirob et al. (2025) [19]. Data were collected from 1,523 patients admitted to Jamalpur Medical College Hospital, Bangladesh, during laboratory-confirmed dengue outbreaks. All data were collected with ethical approval from the Institutional Review Board of Jamalpur Medical College Hospital (approved 10 February 2025). Of the total of 1,523 samples, 1,042 were dengue-positive and 481 were dengue-negative, indicating a moderately unbalanced class distribution. The average age of the patient was 40.3 ± 15.3 years, with an almost equal gender distribution (47% female, 53% male). Each patient record includes 19 hematological and demographic attributes, which are hemoglobin, white blood cell count (WBC), differential counts (neutrophils, lymphocytes, monocytes, eosinophils), platelet count, hematocrit (HCT), mean corpuscular values (MCV, MCH, MCHC), platelet indices (MPV, PDW, PCT) and diagnostic result (positive or negative). Demographic variables age and gender were also recorded to enable population-level analysis.

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3.2. Data Preprocessing

A comprehensive data preprocessing pipeline was designed to ensure consistency, accuracy, and suitability of the hematological dataset for machine learning classification. Initially, the dataset was examined for redundancy, and a total of 12 duplicated records were identified and removed to preserve data integrity. Subsequent analysis confirmed that no missing values were present across any of the 19 hematological and demographic attributes, indicating a complete dataset suitable for downstream modeling.

Outlier detection was performed using the Isolation Forest algorithm [20], an unsupervised ensemble-based anomaly detection method that isolates abnormal observations by constructing random partitioning trees. The anomaly score for a given instance x was computed using Equation 1:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{1}$$

where $E(h(x))$ denotes the average path length of instance x through isolation trees, and $c(n)$ is the average path length of unsuccessful searches in a binary tree containing samples n. Higher values of $s(x, n)$ indicate a stronger probability of anomaly. Using this algorithm, a total of 53 outliers were detected and subsequently handled using Winsorization [21] at the 5th and 95th percentiles to reduce their influence while preserving the overall distribution of hematological values. The Winsorization process is mathematically expressed in Equation 2:

$$x_w = \begin{cases} Q_{0.05}, & \text{if } x < Q_{0.05} \\ x, & \text{if } Q_{0.05} \leq x \leq Q_{0.95} \\ Q_{0.95}, & \text{if } x > Q_{0.95} \end{cases} \tag{2}$$

where $Q_{0.05}$ and $Q_{0.95}$ denote the quantile values of the 5th and 95th percentile, respectively. The effect of Winsorization on representative hematologic parameters is illustrated in Figure 2, showing that extreme values were effectively constrained within a reasonable range while maintaining the overall data distribution.

To ensure uniform feature scaling and improve model convergence, all continuous hematological variables were normalized using z-score standardization, defined in Equation 3:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{3}$$

where x_i represents the feature value, μ is the mean and σ is the standard deviation. This transformation ensured that each variable contributed proportionally to model training and eliminated bias toward attributes with larger numerical ranges.

Since the dataset exhibited moderate class imbalance, with 1,042 dengue-positive and 481 dengue-negative samples, three oversampling strategies were applied and comparatively evaluated: Random Over Sampling (ROS), Random Over Sampling Examples (ROSE), and the Synthetic Minority Oversampling Technique (SMOTE) [22]. The SMOTE algorithm synthetically generates new minority class samples by interpolating between existing minority instances and their nearest neighbors, as shown in Equation 4:

$$x_{new} = x_i + \delta(x_{nn} - x_i) \tag{4}$$

where x_i is a minority class sample, x_{nn} represents one of its nearest neighbors k and δ is a random value within the range [0, 1]. This approach effectively enhances minority representation and mitigates potential overfitting caused by simple duplication of samples.

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After preprocessing, the dataset was free from duplicates and missing values, with outliers properly treated, features standardized, and class imbalance systematically addressed. This refined dataset provided a reliable foundation for subsequent feature selection, model training, and explainability analysis.

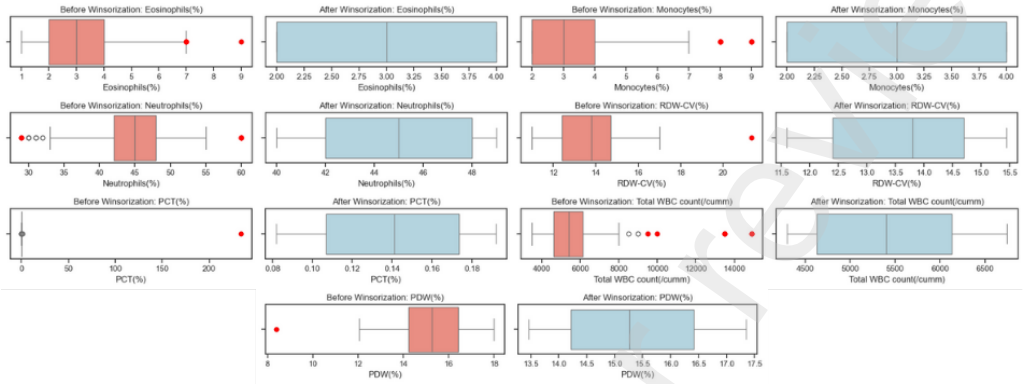


Figure 2: Visualization of outlier treatment using Winsorization at the 5th and 95th percentiles. The box plots illustrate selected hematological parameters (*Eosinophils*, *Monocytes*, *Neutrophils*, *RDW-CV*, *PCT*, *PDW*, and *Total WBC count*) before and after Winsorization. The red-shaded regions represent extreme values identified as outliers, which were adjusted to fall within the specified percentile limits, ensuring reduced skewness while preserving the overall data distribution for robust model training.

3.3. Benchmark Models

To evaluate the performance of the proposed framework, several benchmark machine learning classifiers were used for comparison [23, 24]. These include Support Vector Machine (SVM) [25], Decision Tree (DT) [26], AdaBoost [27], Gradient Boosting (GB) [28], k-Nearest Neighbors (KNN) [29], Histogram-Based Gradient Boosting (HistGB) [30], and Light Gradient Boosting Machine (LGBM) [31]. Each model represents a distinct learning paradigm, enabling a comprehensive assessment of classification robustness and generalization.

The SVM constructs an optimal separation hyperplane that maximizes the margin between the positive and negative classes of dengue. Given a labeled dataset (x_i, y_i), where $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$, the objective of SVM is formulated as:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1 - \xi_i,\ \xi_i \geq 0 \tag{5}$$

where w and b define the hyperplane, ξ_i are slack variables, and C controls the penalty for misclassification.

Decision Trees partition the dataset into homogeneous subsets by recursively selecting features that minimize impurity. The Gini impurity at a node is defined as:

$$G = 1 - \sum_{k=1}^{K} p_k^2 \tag{6}$$

where p_k is the probability of class k at a node. DTs are highly interpretable, but may overfit without pruning.

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Adaptive Boosting (AdaBoost) sequentially combines multiple weak learners (e.g., shallow trees) to form a strong classifier. The final prediction is a weighted sum of individual weak classifiers:

$$F(x) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m h_m(x)\right) \tag{7}$$

where α_m represents the weight assigned to each weak learner $h_m(x)$, based on its classification accuracy.

Gradient Boosting builds an ensemble of decision trees iteratively, where each new learner fits the pseudo-residuals of the previous stage. The model is updated as:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \tag{8}$$

where η is the learning rate that controls the contribution of each weak learner $h_m(x)$.

KNN is a non-parametric classifier that assigns the class of a new sample based on the majority label among its k nearest neighbors. The Euclidean distance is used as the similarity metric:

$$d(x_i, x_j) = \sqrt{\sum_{p=1}^{n} (x_{ip} - x_{jp})^2} \tag{9}$$

KNN is simple and effective for low-dimensional data, but may be computationally expensive for large datasets.

HistGB is an efficient variant of gradient boosting that discretizes continuous features in histograms, significantly improving computational speed and memory usage. It maintains predictive accuracy comparable to conventional gradient boosting while reducing training time through optimized histogram-based split finding.

LGBM is a highly efficient implementation of gradient boosting based on leaf-wise tree growth with depth constraints. It employs histogram-based feature binning and Gradient-based One-Side Sampling (GOSS) to accelerate computation. The objective function is minimized using the loss function gradient:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, F_{m-1}(x_i)) + \lambda \Omega(F_m) \tag{10}$$

where l is the differentiable loss function and $\Omega(F_m)$ is the regularization term to control the complexity of the model.

3.4. Optimizer Framework

To identify the most effective hyperparameter configuration for dengue classification models, multiple optimization strategies were used, including traditional and nature-inspired approaches. Specifically, Grid Search and Optuna were used as baseline optimizers, while three metaheuristic algorithms, which are Genetic Algorithm (GA) [32], Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) [33, 34] that were integrated for advanced global optimization. These optimizers ensure a balance between exploration of the parameter space and exploitation of optimal configurations, minimizing overfitting, and improving classification robustness.

Grid Search is a deterministic optimization approach that exhaustively evaluates all possible combinations of hyperparameters within a predefined search space. Although computationally

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expensive, it guaranties the identification of the best parameter set by minimizing a scoring function.

$$\theta^* = \arg\min_{\theta \in \Theta} \mathcal{L}(f_\theta, D_{val}) \tag{11}$$

where $\mathcal{L}$ represents the validation loss of model f_θ with parameter configuration θ, and D_{val} denotes the validation dataset.

Optuna is an adaptive hyperparameter optimization framework based on Bayesian optimization and the Tree-structured Parzen Estimator (TPE) [35]. It models the probability of achieving a given performance score y conditioned on hyperparameters θ, and iteratively refines the sampling according to the following.

$$\theta^* = \arg\max_{\theta} \frac{l(\theta)}{g(\theta)} \tag{12}$$

where $l(\theta) = P(y < y^*|\theta)$ corresponds to the density of hyperparameters yielding better-than-expected results, and $g(\theta) = P(y \geq y^*|\theta)$ represents the density of poorer configurations. This approach allows for efficient exploration of complex parameter spaces.

The GA mimics natural evolution through selection, crossover, and mutation operations applied to a population of candidate solutions. The optimization objective is expressed as:

$$\max_{\theta} \; F(\theta) = f(\theta) \tag{13}$$

where $F(\theta)$ represents the fitness function (e.g., classification accuracy). A population of candidate solutions N $\{\theta_1, \theta_2, \ldots, \theta_N\}$ is iteratively evolving according to the probability of selection $p_i = F(\theta_i)/\sum_j F(\theta_j)$. The offspring are generated through crossover and mutation to ensure population diversity and avoid premature convergence.

PSO is inspired by the social behavior of bird flocks, where each particle updates its position and velocity based on the best personal and global experiences [36]. For each particle i in a d-dimensional search space, velocity and position are updated as:

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i^t - x_i^t) + c_2 r_2 (g^t - x_i^t) \tag{14}$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{15}$$

where ω is the weight of inertia, c_1 and c_2 are the cognitive and social coefficients, r_1 and r_2 are random numbers in $[0, 1]$, p_i^t denotes the best personal solution, and g^t is the best global solution in iteration t.

GWO emulates the hierarchical leadership and cooperative hunting mechanism of grey wolves [37]. The population is divided into four categories—α, β, δ, and ω—to model leadership dominance. The wolves update their positions according to the leaders' locations as follows:

$$\vec{D} = |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)| \tag{16}$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \tag{17}$$

where $\vec{A} = 2a\vec{r}_1 - a$ and $\vec{C} = 2\vec{r}_2$, with a linearly decreasing from 2 to 0 during iterations, and $\vec{r}_1$, $\vec{r}_2$ being random vectors in $[0, 1]$. The final position $\vec{X}(t+1)$ represents the updated location of the candidate in the search space.

The inclusion of both deterministic (Grid Search, Optuna) and stochastic (GA, PSO, GWO) optimizers ensured a comprehensive exploration of the hyperparameter landscape. Although

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Grid Search and Optuna effectively explored localized regions for fine-tuning, GA, PSO, and GWO facilitated global optimization by avoiding local minima. The best-performing configurations obtained from these methods were subsequently used in the final model evaluation and explanation analysis.

3.5. Proposed Model: GWO-Optimized Extra Trees Classifier (GWO–ETC)

The proposed framework integrates the *Extra Trees Classifier* (ETC) with the *Grey Wolf Optimizer* (GWO) to achieve robust and generalized dengue fever classification using hematological parameters. The GWO algorithm was used to fine-tune the hyperparameters of the ETC by minimizing the classification error on the validation folds, thereby ensuring both high predictive accuracy and stability across different data splits.

The Extra Trees Classifier is an ensemble-based learning algorithm that constructs multiple unpruned decision trees from randomly selected subsets of features and training samples. Each tree contributes to the final prediction by majority vote (for classification) or averaging (for regression). Mathematically, the ETC ensemble can be expressed as:

$$\hat{y}(x) = \text{mode}\{h_t(x) \mid t = 1, 2, \ldots, T\} \tag{18}$$

where $\hat{y}(x)$ denotes the predicted class label, T is the total number of trees in the ensemble, and $h_t(x)$ represents the prediction of the decision tree t^{th}. Each tree is trained using a randomly selected subset of features $F_t \subseteq F$ and samples $S_t \subseteq S$, increasing the diversity of the model and reducing variance.

To optimize ETC parameters, the Grey Wolf Optimizer (GWO) guided the search for an optimal parameter vector

$\Theta = \{n_{\text{estimators}}, \text{max_depth}, \text{min_samples_split}, \text{min_samples_leaf}, \text{max_features}, \text{class_weight}\}$

that minimizes the classification loss function. The optimization objective is defined as:

$$\min_{\Theta} \mathcal{L}_{\text{ETC}}(\Theta) = 1 - \text{Accuracy}(\Theta) \tag{19}$$

where $\mathcal{L}_{\text{ETC}}(\Theta)$ represents the loss corresponding to a particular hyperparameter configuration Θ. The GWO algorithm iteratively updates the candidate solutions based on the hunting behavior of grey wolves, as described by Equations 16 and 17. At each iteration, the position vectors (α, β, δ) represent the top-performing configurations, and the rest of the wolves (ω) update their positions according to the leaders' guidance until convergence.

The optimization process yielded the following optimal hyperparameters for the ETC model:

Table 2: Optimized hyperparameters of the GWO–ETC model.

Hyperparameter	**Optimized Value**
Number of estimators ($n_{\text{estimators}}$)	298
Maximum depth (max_depth)	20
Minimum samples split (min_samples_split)	3
Minimum samples leaf (min_samples_leaf)	1
Maximum features (max_features)	None
Class weight (class_weight)	None

The final prediction function of the GWO-optimized model is given by:

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$$\hat{y}^* = \text{mode}\{h_t(x;\Theta^*) \mid t = 1, 2, \ldots, 298\} \tag{20}$$

where $h_t(x;\Theta^*)$ denotes the decision function of the optimized tree t^{th} trained under the best parameter configuration Θ^*. The proposed GWO–ETC model combines the high variance-reduction capability of ensemble methods with the global search efficiency of metaheuristic optimization, resulting in improved classification accuracy, interpretability, and generalization for dengue fever detection.

3.6. Evaluation Metrics

The performance of the proposed GWO–ETC model was evaluated using a range of statistical and graphical metrics to ensure reliability and clinical relevance. Accuracy measured the general accuracy of the predictions, while precision, recall, and F1-score assessed the sensitivity and balance of the model between dengue-positive and dengue-negative classifications. Matthews'correlation coefficient (MCC) and Cohen's Kappa were included to provide robust measures of agreement under class imbalance, and the confusion matrix was examined to identify specific misclassification patterns. Graphical assessments further explained the predictive behavior of the model. The Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves evaluated discrimination ability, the calibration graph assessed probability reliability, and the lift and entropy charts measured case prioritization and prediction uncertainty [38, 39]. Collectively, these metrics confirmed that the GWO–ETC model achieved high predictive accuracy, reliable calibration, and strong generalization in dengue fever classification.

3.7. Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) techniques, namely LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) [40, 41], were employed to enhance the transparency and interpretability of the predictive framework. The SHAP method quantifies the contribution of each feature to the model output by computing Shapley values derived from cooperative game theory, providing a global interpretation of feature influence in dengue severity prediction. The contribution of a feature i is expressed by the Shapley value formulation shown in Equation 21.

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \tag{21}$$

Here, N represents the set of all features, S is a subset of features excluding i, and $f(S)$ denotes the model output when considering the subset S. This formulation ensures an equitable distribution of each feature's contribution to the final prediction.

In contrast, LIME provides local interpretability by approximating the behavior of the complex model with a simpler, interpretable surrogate model around a specific instance. It minimizes the objective function presented in Equation 22.

$$\xi(x) = \arg\min_{g \in G} \ \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{22}$$

In Equation 22, f denotes the original model, g represents an interpretable model belonging to the family of models G, $\mathcal{L}$ measures the fidelity between f and g weighted by the locality kernel π_x, and $\Omega(g)$ controls the complexity of g.

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3.8. Experimental Setup

All experiments were conducted on a workstation running Ubuntu 22.04 LTS equipped with an Intel Core i5-11400H processor (6 cores, 12 threads), 16 GB of DDR4 RAM operating at 3200 MHz, and an NVIDIA GeForce RTX 3050 GPU with 4 GB of dedicated memory. The experimental environment used Python version 3.10.1. Model training, hyperparameter optimization, and evaluation were performed under a stratified cross-validation scheme to ensure robustness and fairness across all models. Random seeds were fixed throughout the workflow to maintain the reproducibility of the results.

4. Results and Discussion

This section presents the experimental results obtained from the benchmark machine learning models and the proposed GWO–ETC framework for dengue fever classification using hematological features. All models were evaluated under four class-balancing strategies—without resampling (W/o), Random Over Sampling (ROS), Random Over Sampling Examples (ROSE) and Synthetic Minority Oversampling Technique (SMOTE)—to analyze their performance stability under class imbalance conditions. Each model was assessed based on multiple evaluation metrics, including accuracy, precision, recall, F1-score, Cohen's Kappa and Matthews Correlation Coefficient (MCC), providing a holistic view of predictive capability, robustness and reliability.

Table 3 summarizes the performance of all classifiers under different data balancing methods. In general, the ensemble-based models demonstrated superior predictive capability compared to traditional learners across all evaluation metrics. Among these, the Extra Trees Classifier (ETC) with Random Over Sampling (ROS) produced the highest overall performance, achieving an accuracy, precision, recall and F1-score of 0.9227, accompanied by a Cohen's Kappa of 0.8454 and an MCC of 0.8464, reflecting strong classification reliability and inter-rater agreement. Random Forest (RF) and Histogram-based Gradient Boosting (HistGB) also performed competitively under the ROS method, achieving accuracies greater than 0.89, confirming the robustness of ensemble learning for the classification of hematological data.

The comparatively lower performance of models such as SVM, Decision Tree (DT), and K-Nearest Neighbors (KNN) can be attributed to their limited capacity to model complex, nonlinear feature interactions and their sensitivity to class imbalance. In contrast, ensemble models combine multiple weak learners, capturing diverse feature relationships and reducing overfitting through averaging or boosting. The superior results of ETC, particularly under ROS, arise from its use of random feature splits and de-correlated tree ensembles, enabling it to capture subtle diagnostic variations between dengue-positive and negative samples. Moreover, the ROS technique improved minority class representation without introducing synthetic noise, allowing the model to learn balanced decision boundaries. These outcomes suggest that ensemble learners with stochastic regularization offer higher generalization power and diagnostic reliability in biomedical datasets characterized by overlapping class distributions and moderate imbalance.

After balancing the dataset using Random Over Sampling (ROS), the Extra Trees Classifier (ETC) was optimized using five different optimization strategies: Optuna, Grid Search, Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Grey Wolf Optimizer (GWO). Each optimizer fine-tuned the model's hyperparameters to enhance classification performance and generalization. Table 4 presents the comparative results based on six evaluation metrics: Accuracy, Precision, Recall, F1-score, Cohen's Kappa, and Matthews Correlation Coefficient (MCC).

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Table 3: Performance comparison of benchmark classifiers under different data balancing methods. The best results for each model are highlighted in bold.

Model	Balancing	Accuracy	Precision	Recall	F1 Score	Cohen Kappa	MCC
SVM	W/o	0.7162	0.6660	0.6046	0.6083	0.2408	0.2635
	ROS	0.6425	0.6436	0.6425	0.6418	0.2850	0.2861
	ROSE	0.6208	0.6223	0.6208	0.6196	0.2415	0.2431
	SMOTE	0.6377	0.6381	0.6377	0.6374	0.2754	0.2758
DT	W/o	0.6799	0.6303	0.6324	0.6313	0.2626	0.2627
	ROS	0.8261	0.8560	0.8261	0.8224	0.6522	0.6814
	ROSE	0.7681	0.7681	0.7681	0.7681	0.5362	0.5362
	SMOTE	0.7198	0.7219	0.7198	0.7192	0.4396	0.4417
KNN	W/o	0.6403	0.5489	0.5378	0.5353	0.0830	0.0860
	ROS	0.6256	0.6344	0.6256	0.6194	0.2512	0.2599
	ROSE	0.6449	0.6450	0.6449	0.6449	0.2899	0.2899
	SMOTE	0.6763	0.6839	0.6763	0.6730	0.3527	0.3601
AdaBoost	W/o	0.6964	0.6302	0.5873	0.5888	0.1984	0.2132
	ROS	0.6401	0.6408	0.6401	0.6396	0.2802	0.2809
	ROSE	0.8019	0.8093	0.8019	0.8007	0.6039	0.6112
	SMOTE	0.7512	0.7519	0.7512	0.7510	0.5024	0.5031
GB	W/o	0.7525	0.7571	0.6310	0.6388	0.3130	0.3670
	ROS	0.7729	0.7730	0.7729	0.7729	0.5459	0.5459
	ROSE	0.8261	0.8415	0.8261	0.8241	0.6522	0.6674
	SMOTE	0.8116	0.8237	0.8116	0.8098	0.6232	0.6352
LGBM	W/o	0.7459	0.7222	0.6376	0.6476	0.3180	0.3497
	ROS	0.8357	0.8548	0.8357	0.8335	0.6715	0.6903
	ROSE	0.7850	0.7851	0.7850	0.7850	0.5700	0.5701
	SMOTE	0.7705	0.7713	0.7705	0.7704	0.5411	0.5418
HistGB	W/o	0.7327	0.7405	0.5937	0.5869	0.2318	0.3002
	ROS	0.8937	0.9000	0.8937	0.8933	0.7874	0.7937
	ROSE	0.8213	0.8337	0.8213	0.8196	0.6425	0.6549
	SMOTE	0.8213	0.8237	0.8213	0.8209	0.6425	0.6450
RF	W/o	0.7558	0.7678	0.6334	0.6416	0.3198	0.3780
	ROS	0.8986	0.9023	0.8986	0.8983	0.7971	0.8008
	ROSE	0.8188	0.8306	0.8188	0.8172	0.6377	0.6493
	SMOTE	0.7971	0.8011	0.7971	0.7964	0.5942	0.5982
ETC	W/o	0.7690	0.7986	0.6487	0.6610	0.3566	0.4215
	ROS	**0.9227**	**0.9237**	**0.9227**	**0.9227**	**0.8454**	**0.8464**
	ROSE	0.8986	0.8991	0.8986	0.8985	0.7971	0.7977
	SMOTE	0.8285	0.8302	0.8285	0.8283	0.6570	0.6587

Table 4: Performance comparison of ETC under different optimization techniques after ROS balancing. The best results are highlighted in bold.

Optimizer	Accuracy	Precision	Recall	F1 Score	Cohen Kappa	MCC
Optuna	0.9444	0.9500	0.9444	0.9443	0.8889	0.8944
Grid Search	0.9306	0.9309	0.9306	0.9305	0.8611	0.8614
PSO	0.9306	0.9390	0.9306	0.9302	0.8611	0.8695
GA	0.9453	0.9412	0.9380	0.9395	0.9105	0.9112
GWO	**0.9578**	**0.9541**	**0.9516**	**0.9524**	**0.9254**	**0.9260**

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As shown in Table 4, the Grey Wolf Optimizer (GWO) achieved the best overall performance, outperforming all other optimization strategies across every metric. GWO achieved a precision of 0.9578, a precision of 0.9541, a recall of 0.9516, and an F1-score of 0.9524, along with the highest Cohen's Kappa (0.9254) and MCC (0.9260) values. These results demonstrate GWO's superior global search capability and convergence stability, enabling effective exploration of the hyperparameter space and avoidance of local minima. In comparison, Optuna and GA also produced competitive results, achieving accuracy levels above 0.94, whereas Grid Search and PSO showed slightly lower performance due to their deterministic or parameter-sensitive search strategies. The overall result validates the effectiveness of nature-inspired optimization in the refinement of ensemble models, with GWO providing the most balanced and robust configuration for the classification of dengue fever.

4.1. Proposed Model Result Analysis

The performance of the proposed GWO–ETC model was further analyzed through multiple evaluation perspectives to ensure diagnostic reliability and generalizability. Figure 3a presents the confusion matrix, illustrating the model's ability to accurately differentiate between dengue-positive and dengue-negative cases. Of 414 total samples, 197 cases positive for dengue and 198 cases negative for dengue were correctly classified, while only 19 cases were misclassified. This near-perfect prediction pattern reflects the balanced sensitivity and specificity of the model, aligning with its high recall (0.9516) and precision (0.9541) values. The minimal false negatives demonstrate the strong ability of the model to identify actual dengue cases, which is clinically vital in early diagnosis.

The ROC and Precision–Recall curves shown in Figures(3b) and (3c) further confirm the robust discriminative performance of the model. The ROC curve achieved an area under the curve (AUC) of 0.9875, indicating excellent separation between classes with minimal overlap. Similarly, the Precision–Recall curve obtained an AUC of 0.9811, demonstrating high precision even at elevated recall thresholds, emphasizing the model's ability to maintain accuracy within different decision boundaries. These findings highlight the superior stability and predictive consistency of the GWO–ETC model.

The lift chart (Figure (3d)) provides additional information about the ranking capability of the model. The GWO–ETC model achieved a maximum lift value of 2.0, indicating that the highest predicted samples are twice as likely to be true dengue-positive cases compared to random selection. This shows the effectiveness of the model for prioritizing high-risk patients, which is crucial for clinical triage and surveillance.

Collectively, these graphical evaluations affirm that the GWO–ETC framework provides strong predictive performance, balanced classification, and consistent probability calibration. Its high AUC scores, strong lift ratio and low entropy variance substantiate the robustness and reliability of the model for practical deployment in dengue diagnosis based on hematological features.

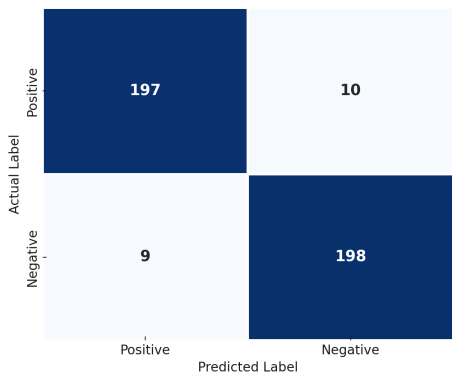
4.2. Explainability Analysis

To enhance model transparency and provide clinical interpretability, Explainable Artificial Intelligence (XAI) techniques were employed using SHAP and LIME. These methods provided global and local information on how the hematological parameters influenced the classification of dengue-positive and dengue-negative cases using the proposed GWO–ETC model.

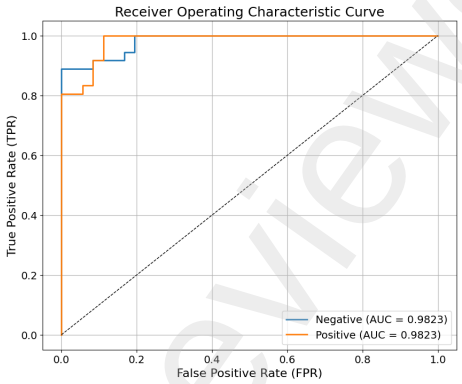
SHAP (SHapley Additive exPlanations) was used to quantify the contribution of each feature to the model's output. The global SHAP summary plot, shown in Figure 4(a), identifies the most

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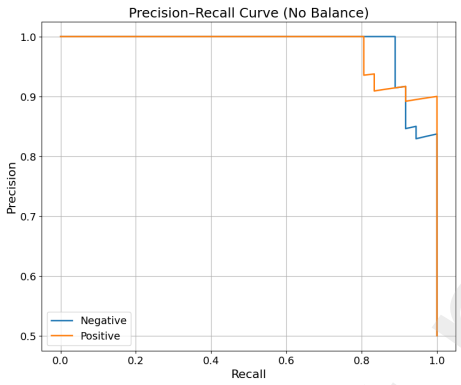
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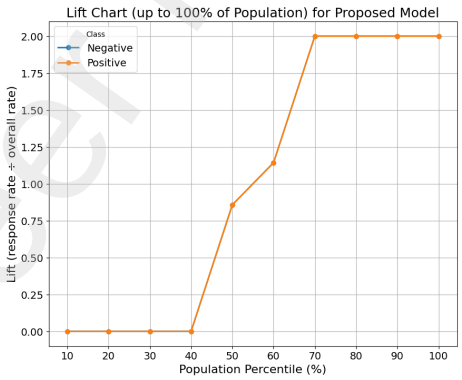
(a) Confusion Matrix



(b) ROC Curve (AUC = 0.9875)



(c) Precision–Recall Curve (AUC = 0.9811)



(d) Lift Chart

Figure 3: Graphical evaluation of the proposed GWO–ETC model on the balanced dengue dataset. (a) Confusion matrix illustrating classification accuracy and misclassifications, (b) ROC curve demonstrating excellent class separability (AUC = 0.9875), (c) Precision–Recall curve confirming high precision and recall (AUC = 0.9811), and (d) Lift chart showing the model's ability to effectively prioritize dengue-positive cases.

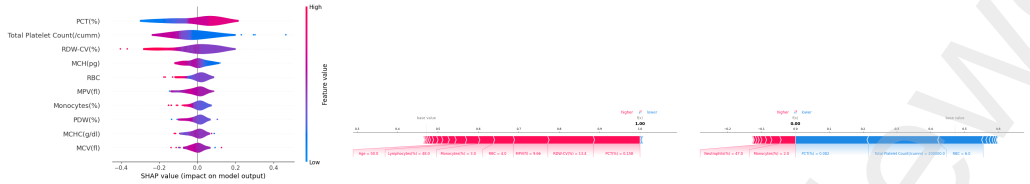
influential features that determine the predictions of the model. Platelet count, Plateletcrit (PCT), RDW-CV, and RBC emerged as the top-ranking determinants. Higher platelet and PCT values were associated with dengue-negative predictions, whereas lower platelet and higher RDW-CV values pushed the prediction towards the dengue-positive class. These findings align with known hematological indicators of dengue, confirming that platelet depletion and hematocrit variability are critical diagnostic markers.

The SHAP force plots in Figures 4(b) and 4(c) illustrate local feature contributions for individual predictions. In the positive sample, attributes such as elevated lymphocytes, reduced monocytes, and low RBC values strongly contributed to the classification of dengue-positive. In contrast, in the negative case, a higher platelet count and normal PCT values drove the prediction towards the dengue-negative class. This consistency between SHAP outputs and clinical reasoning validates the biological relevance of the model's decision process.

To complement SHAP explanations, Local Interpretable Model-Agnostic Explanations (LIME) were applied to provide human-understandable local approximations of model predictions. The

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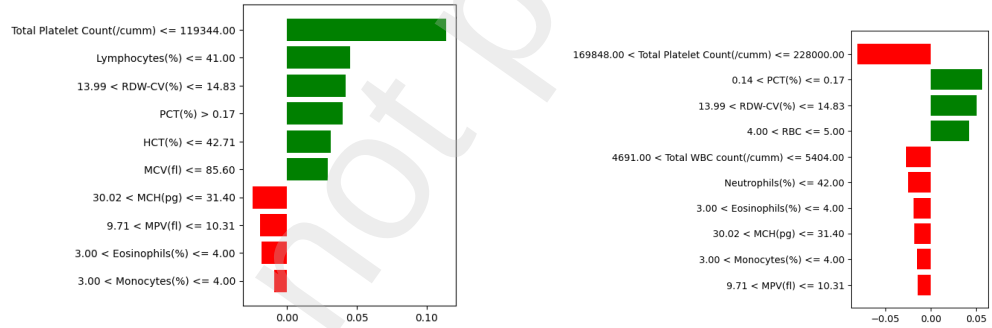


(a) Global SHAP summary plot (b) SHAP force plot (positive sample) (c) SHAP force plot (negative sample)

Figure 4: SHAP-based interpretation of the GWO–ETC model. (a) Global SHAP summary highlights the top hematological features contributing to dengue prediction. (b) Force plot for a dengue-positive instance showing feature contributions towards the positive class. (c) Force plot for a dengue-negative instance illustrating features contributing to non-dengue classification.

LIME explanations for the representative dengue-positive and dengue-negative samples are presented in Figure 5. For dengue-positive predictions, low platelet count, reduced mean platelet volume (MPV) and low PCT were identified as the main contributing factors, reinforcing their diagnostic importance. In contrast, the dengue-negative sample was mainly influenced by higher platelet levels, normal hematocrit (HCT), and mean corpuscular volume (MCV), reflecting physiologically stable conditions.

The alignment between SHAP and LIME interpretations strengthens confidence in the explainability of the GWO–ETC model. Both methods consistently emphasized platelet-related and red blood cell–related features as key determinants of classification, providing a clear and clinically meaningful justification for the model's output.



(a) LIME explanation (positive case) (b) LIME explanation (negative case)

Figure 5: LIME-based local interpretation of the GWO–ETC model. (a) Explanation for a dengue-positive instance highlighting low platelet and PCT levels as the strongest predictors. (b) Explanation for a dengue-negative instance showing that higher platelet and hematocrit values contributed most to the non-dengue prediction.

4.3. Ablation Study

To evaluate the relative contribution of each component in the proposed framework, an ablation study was conducted. The analysis systematically examined how performance changed when specific elements—data balance, optimization, and integration of explainability, integration—were modified or removed. Table 5 presents the comparative results, highlighting the incremental impact of each design choice on the diagnostic capability of the model.

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Initially, the baseline Extra Trees Classifier (ETC) was trained on the original imbalanced dataset without optimization, achieving an accuracy of 0.7690 and an F1-score of 0.6610. Incorporating Random Over Sampling (ROS) significantly improved sensitivity towards dengue-positive cases, increasing accuracy to 0.9227 and F1-score to 0.9227, demonstrating the effectiveness of data balancing in mitigating class bias. When optimization techniques were introduced, the performance improved further across all metrics. Deterministic optimizers such as Grid Search and Optuna offered moderate gains; however, nature-inspired algorithms provided superior results due to their ability to perform global exploration within the hyperparameter space. Among them, the Grey Wolf Optimizer (GWO) achieved the highest overall performance, with accuracy of 0.9578, precision of 0.9541, recall of 0.9516, and F1-score of 0.9524.

To assess the role of model explainability, the optimized GWO–ETC model was subsequently integrated with the SHAP and LIME frameworks. Although these XAI modules did not alter the numerical accuracy, they enhanced transparency and interpretability highlighting essential criteria for clinical deployment. The SHAP and LIME analyzes confirmed that platelet count, platelet crit (PCT), RDW-CV, and RBC were consistently identified as key determinants, validating the biological plausibility of the predictions of the model.

In general, the ablation results demonstrate that each module ROS balancing, GWO-based optimization, and XAI integration contributes to improved robustness, precision, and clinical interpretability of the final model. Thus, the complete GWO–ETC framework provides an optimal balance between predictive performance and explainable decision-making for dengue diagnosis based on hematological features.

Table 5: Ablation study showing the impact of different components on model performance.

Model Variant	**Accuracy**	**Precision**	**Recall**	**F1-Score**
ETC (Baseline, no balancing)	0.7690	0.7986	0.6487	0.6610
ETC + ROS (Balanced)	0.9227	0.9237	0.9227	0.9227
ETC + ROS + GWO (Proposed)	**0.9578**	**0.9541**	**0.9516**	**0.9524**

5. Conclusion

This study proposed an explainable learning framework inspired by nature for dengue fever classification based on hematological parameters. The optimized GWO–ETC model achieved a high classification accuracy of 0.9578, with precision, recall, and F1-score exceeding 0.95, underscoring its diagnostic reliability and stability. The integration of the Grey Wolf Optimizer (GWO) effectively enhanced hyperparameter tuning, leading to improved convergence and reduced overfitting. Explainable AI analysis using SHAP and LIME provided valuable insight into the internal decision logic of the model, identifying platelet count, plateletcrit (PCT), red cell distribution width (RDW-CV), red blood cell count (RBC), lymphocytes and monocytes as the most influential predictors. These hematological biomarkers are clinically recognized indicators of dengue severity, and their consistent prominence in global and local interpretability analyzes reinforces the medical validity of the model's predictions. The alignment between computational reasoning and established clinical evidence confirms the proposed GWO–ETC framework as an accurate and reliable diagnostic aid.

Despite its strong performance, this study has certain limitations. The dataset originated from a single regional source, which may not capture population-level variations in hematological responses to dengue. In addition, only routine blood parameters were analyzed, excluding

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virological, immunological, and biochemical markers that could improve prediction granularity. Future studies will focus on expanding the dataset across multiple hospitals and integrating multimodal features such as serological test results, liver function markers, and environmental risk factors. Further research will also explore advanced hybrid metaheuristic optimizers and explainable frameworks based on deep learning to improve model generalization and real-time clinical deployment. Longitudinal data collection could enable disease severity staging and prognostic modeling, extending the clinical utility of this framework.

The findings of this study are of strong clinical relevance. Using routinely available hematological tests, the proposed model offers a rapid, cost-effective, and interpretable diagnostic support tool for early detection of dengue. The identification of platelet count, PCT, RDW-CV, and RBC as key predictors aligns with the clinical understanding of thrombocytopenia and hematocrit fluctuation in dengue progression. The inclusion of lymphocytes and monocytes in the predictive hierarchy reflects the characteristic immune response of viral infections, providing a physiologically coherent basis for classification. Through SHAP and LIME explanations, clinicians can visualize the contribution of each feature for individual patients, promoting transparency and confidence in automated predictions.

Data Availability

The datasets produced or analyzed in this study can be obtained from the corresponding author upon request, subject to reasonable terms.

Conflict of Interest Statement

The authors declare no conflict of interest.

Ethical Approval

Not applicable.

Informed Consent

Not applicable.

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